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CHM 211 — Past Questions & Materials
by Renance

Course: CHM 211
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200

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Paper 1 of 4 — CHM 211 Organic Chemistry 1

Level 200 • First Semester • PastQuestion

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UNIVERSITY OF PETROLEUM RESOURCES, E |

_ DEPARTMENT OF CHEMISTRY |

FIRST SEMESTER EXAMINATION, 2024/2025 SESSION, |

CHM 211 ORGANIC CHEMISTRY 1

Answer Any Four Questions. Time Allowed: 2hours.

Group the following compounds into (1) Aromatic (ii) Anti-aromatic and Gn) non-aromatic. Support your answer with proof or calculauon where applicable (!0 mks).

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la}

1b) List four methods of preparing cycloalkanes (4 mks).

ic) Complete the reaction below (3 marks)

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0 7

i ae a [>

/ A

ee z

2a Define Aromaticity and state the three conditions for aromaticity (4 mks}.

2b) Name the following compounds: (4 mks)

- - .

Br

2c} Briefly expiain the following

- (i) Monosaccharides (ii} Disaccharides. (mks)

2d) Differentiaic between Amylose and Amylopec in. = (2mks}

Ze) Draw the structure of sucrose. (2mbks}

21) Diferentiate an alduse from a heiose sugai (¢ amb)

a6 py; - H - a

a) Brierly explain the followimg(). Giveosidie bond (1) Peptide bona (ii) Essential amino acid

iv) Non-essenta! amine acid (Qmks)

i . - " € - D glu os to D-gluct ! the fete wing reaction was cervied out Give the missin

nis tabcied as A, Band C, 7 MEE SOULE

CH2OH

HO—H A.B wane -OH

H— OH —i A

H—OH - H~ 4 OH

CH₂OH H— OF

CH₂OH (3mks)

Draw the structures of the following compounds

(i) D-ribose (ii) Aldotriose (iii) Ketotriose. (3mks)

3d) Using Harworth projection. draw the structure of D-glucose. (2mks)

3e) | what are the main groups in an amino acid? (1 mks)

4a Draw the general structure of the transition state of S_N2 reaction (1'2mks)

4b In the reaction $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{Alc KOH} \xrightarrow{*} \text{CH}_3\text{CH}=\text{CH}_2 + \text{KBr}$, show how use of isotopes can contribute to determining the mechanism (4mks)

4c Give an equation for the fission of the bond between carbon and bromine in the compound $\text{CH}_3\text{CH}_2\text{Br}$ to form an electrophile. Use arrow to show the electron movement. (3mks)

4d Write an S_N2 mechanism for the reaction $\text{C}_6\text{H}_5\text{CH}_2\text{Br} + \text{KOH} \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{KBr}$ (5mks)

4e Explain how nature of alkyl groups can affect the rate of reaction of S_N2 mechanism of R - Br (4marks)

5a Using a table, differentiate homolytic fission from a heterolytic fission (7 marks)

5b What isomerism is there between propanone and propanal (mk).

5c: The hydrolysis of cyanohydrins produces ----- ? Show the process. (2mks)

i

5d. What is the simplest keto carboxylic acid? How can it be synthesized and give one use of it? (4mks)

5e. With the aid of a chemical reaction outline the synthesis of any α,β-unsaturated acid. (2mks)

5f What is the effect of the chloro group on the acidity of $\text{CH}_3\text{CHClCOOH}$ compared to

$\text{CH}_3\text{CH}_2\text{COOH}$ (1'4mks)

With the aid of chemical equations show how dicarboxylic acids can be synthesized from cyclic ketones and by cyanide synthesis. (3mks)

6b. Illustrate with chemical equations two reactions of hydroxy carboxylic acid. (2'0mks)

6c. Give two uses of oxalic acid (2mks)

6d. Hell-Volhard-Zelinsky describes the synthesis of what type of halogeno acid. Explain the synthesis.

(3mks,

c. What is the basic property of a hydroxy acid? (1mk)

or. With the aid of chemical equations only. explain the step-by-step process of synthesizing a β-keto acid. (4mks)

acid. (4mks)

6g. Draw and write the names of any two dicarboxylic acid of your choice. (2mks)

END OF PAPER

Paper 2 of 4 — CHM 211 ORGANIC CHEMISTRY 1

Level 200 • First Semester • PastQuestion

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UNIVERSITY OF PETROLEUM

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FIRST SEMESTER EXAMIN

CHM 211 ORGANIC CHEMISTRY !

Time Allowed: 1 hour.

Answer Any Four Questions |

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-aromatic and (11)

lowing compounds into (1) Aromatic Py sae

proc! or calculation where applicable

your answers with

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aring cycloalkanes (4 mks).

1a) Group the ite

1b) List four methods of prep

1c) Complete the reaction below (3 marks)

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— ry (4 mks).

2a) Define Aromaticity and state the three conditions for aromaticity

2b) Name the following —e (4 mks)

PASE

2c} Briefly explain the follow1 a

- (1) Monosaccharides (11) Disaccharides. (3mks)

2d) Differentiate between Amylose and Amylopectin. (2mks) °

2e) Draw the structure of sucrose. (2mks)

2i) Differentiate an aldose from a ketose sugar (1 2mk)

3a) Briefly explain the following(i). Glycosidic b

ond (11) Peptide bond Gii} Essential 2

(iv) Non-essential amino acid (mks) Gl) Essential amino acid

3b) In the STIVE Ss ion sftD

4 -glucose to D- uci, | the following reacti:

ie is labeled as A, B and C, i.) and the EBON Wag carried out. Give the missing

Mes < oie samt

H—~ :

OH Cc H— OH

CH₂OH Hop

Cc

2OH (3mks)

3c) Draw the structures of the following compounds

(1) D-ribose (ii) Aldotriose (iii) Ketotriose. (3mks)

3d) Using Harworth projection, draw the structure of D-glucose. (2mks)

3e) what are the main groups in an amino acid? (1% mks)

4a Draw the general structure of the transition state of S_N2 reaction (1'2mks)

4b [In the reaction $\text{CH}_3\text{CH}=\text{CH}_2\text{Br} \xrightarrow{\text{Alc KOH}} \text{CH}_3\text{CH}=\text{CH}_2 + \text{KBr}$, show how use of isotopes can contribute to determining the mechanism (4mks)

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5e. With the aid of a chemical reaction outline the synthesis of any

5f What is the effect of the chloro group on the acidity of $\text{CH}_3\text{CHClCOOH}$ compared to $\text{CH}_3\text{CH}_2\text{COOH}$ (1% mks)

6a. With the aid of chemical equations show how dicarboxylic acids can be synthesized from cyclic ketones and by cyanide synthesis. (3mks)

f hydroxy! carboxy] acid. (2'amks)

6b. illustrate with chemical equations two reactions of

6c. Give two uses of oxalic acid (2mks)

6d. Hell-Volhard-Zelinsky describes the synthesis of what type of halogeno acid? Explain the synthesis.

6f. What is the basic property of a hydroxyl acid? (1mk)

6r. With the aid of chemical equations only, explain the step-by-step process of synthesizing a β -keto acid.

(4mks)

Draw and write the names of any two dicarboxylic acids of your choice. (2mks)

be.

END OF PAPER

Paper 3 of 4 — CHM 211 OXIDATION AND REDUCTION REACTION

Level 200 • Second Semester • Handout

This source paper is a scanned document; its pages are reproduced below.

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OXIDATION AND REDUCTION REACTION

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✓ Oxidation: Refers to gaining oxygen, losing hydrogen or losing one or more electrons.

✓ Reduction: Refers to losing oxygen, gaining hydrogen or gaining one or more electrons.

Oxidation and Reduction steps Complement one another

e.g. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

$\downarrow$
Oxidation

$\uparrow$
Reduction

Magnesium is oxidized, while oxygen is reduced. Magnesium acts as the reducing agent or reductant, while O_2 acts as the oxidizing agent or oxidant.

$\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$ — Oxidation.

$\text{O}_2 + 4\text{e}^- \rightarrow 2\text{O}^{2-}$ — Reduction.

Exercises

A Which species is being oxidized, & which is reduced in each of the fll rxns or half rxns?

- 1 $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$
- 2 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- 3 $2\text{Ag}^+ + \text{Zn} \rightarrow 2\text{Ag} + \text{Zn}^{2+}$
- 4 $\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$
- 5 $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
- 6 $\text{Mg} + 2\text{H}^+ \rightarrow \text{Mg}^{2+} + \text{H}_2$

B Classify each of the fll changes as a reduction process, an oxidation process, a redox rxn or change that does not involve reduction or oxidation.

- ① $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
- ② $[\text{MnO}_4]^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$
- ③ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

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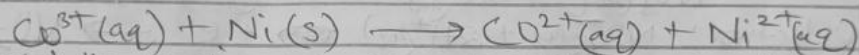
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- ④ $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- ⑤ $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$
- ⑥ $\text{H}_2\text{O} + \text{Cl}_2 \rightarrow \text{HCl} + \text{HOCl}$
- ⑦ $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
- ⑧ $\text{Mg} + 2\text{HNO}_3 \rightarrow \text{Mg}(\text{NO}_3)_2 + \text{H}_2$

Balancing Redox Equations.

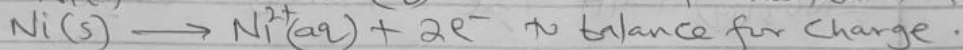
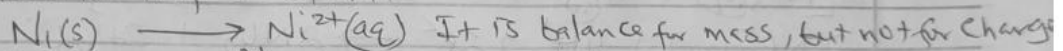
Redox reactions are reactions that involve the transfer of electrons between chemical species.

The Half-reaction method of Balancing redox reactions. For example consider the rxn btw the CO^{3+} ion and nickel metal:



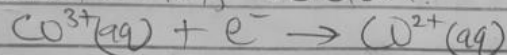
Split the equation into separate Oxidation and Reduction half-rxns!

Oxidation half-reaction:

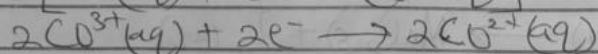
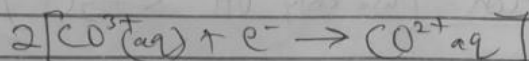


Net charge on both side is zero.

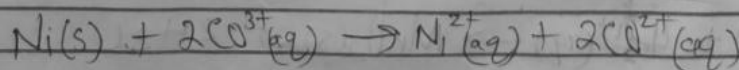
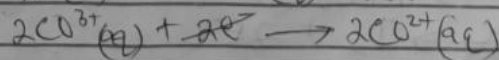
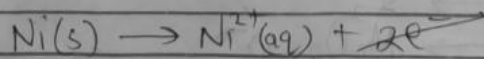
Reduction half-reaction:



Combining the two half-reactions together!



Add up.



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Balancing Redox equations in acidic or basic solution.
For acidic medium we may need to add H_2O or H^+ ions or OH^- ions for reactions in basic solution.

Example 1: Redox in acidic solution.

$$Cu(s) + NO_3^-(aq) \rightarrow Cu^{2+}(aq) + NO_2$$

Oxidation: $Cu(s) \rightarrow Cu^{2+}(aq)$
Reduction: $NO_3^-(aq) \rightarrow NO_2(g)$
Balancing for mass and charge

Oxidation: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ — (1)
Reduction: $NO_3^-(aq) \rightarrow NO_2(g) + H_2O$
 $NO_3^-(aq) + 2H^+(aq) \rightarrow NO_2(g) + H_2O$
Reduction: $NO_3^-(aq) + 2H^+(aq) + e^- \rightarrow NO_2$ — (2)

$$2[NO_3^-(aq) + 2H^+(aq) + e^- \rightarrow NO_2 + H_2O]$$

$$2NO_3^-(aq) + 4H^+ + 2e^- \rightarrow 2NO_2 + 2H_2O$$

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

$$Cu(s) + 2NO_3^-(aq) + 4H^+ \rightarrow Cu^{2+}(aq) + 2NO_2 + 2H_2O$$

Example 2: For Basic solution (use only OH^- & H_2O)

$$MnO_4^-(aq) + I^-(aq) \rightarrow MnO_2(s) + I_2(aq)$$

Oxidation: $I^-(aq) \rightarrow I_2$
Reduction: $MnO_4^-(aq) \rightarrow MnO_2(s)$
Oxidation: $2I^-(aq) \rightarrow I_2 + 2e^-$

$$MnO_4^-(aq) + 4H^+(aq) \rightarrow MnO_2(s) + 2H_2O$$

$$MnO_4^-(aq) + 4H^+(aq) + 4OH^-(aq) \rightarrow MnO_2(s) + 2H_2O + 4OH^-(aq)$$

$$+ H_2O(l)$$

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QUESTION NO.

$\text{MnO}_4^- (\text{aq}) + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2 (\text{s}) + 4\text{OH}^- (\text{aq})$

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Paper 4 of 4 — CHM 211 Organic Chemistry I NOTES - BOHR THEORY

Level 200 • First Semester • Handout

CHM 211 NOTES

THE BOHR'S MODEL OF THE ATOM

1.0 OBJECTIVES

At the end of this unit you should be able to:

- Describe the components of the atom
- Describe the principles of atomic excitation and de-excitation
- State the origin of the spectral lines in the hydrogen spectrum
- Explain the postulates which altogether make up the Bohr Theory.
- Identify the various line series in the hydrogen spectrum
- State the shortcomings of the Bohr Theory

1.1 Composition of the atom

Atoms have a definite structure. This structure determines the chemical and physical properties of matter. This atomic structure was not fully understood until the discovery of the neutron in 1932. The history of the discovery of atomic structure is one of the most interesting and profound stories in science. In 1910 Rutherford was the first to propose what is accepted today as the basic structure of the atom. Today the Rutherford model is called the "planetary" model of the atom. In the planetary model of the atom there exists a nucleus at the center made up of positively charged particles called "protons" and electrically neutral atoms called "neutrons". Surrounding or "orbiting" this nucleus are the electrons. In elements the number of electrons equals the number of protons.

Figure 1.1: Composition of the atom

The picture above greatly exaggerates the size of the nucleus relative to that of the atom. The nucleus is about 100,000 times smaller than the atom. Nevertheless, the nucleus contains essentially all of the mass of the atom. In order to discuss the mass of an atom we need to define a new unit of mass appropriate to that of an atom. This new unit of mass is called the "atomic mass unit" or amu. The conversion between the amu and gram is:

$$1 \text{ amu} = 1.67 \times 10^{-24} \text{ g}$$

The mass, in amu, of the three particles is given in the table below:

1

Table 1.1: Properties of the sub-atomic particles

Name	Symbol	Electrical Charge	Mass (amu)
------	--------	-------------------	------------

Electron	e	-1	0.000549
----------	---	----	----------

Proton	P	+1	1.00728
--------	---	----	---------

Neutron	n	0	1.00867
---------	---	---	---------

Note that the mass of an electron is about 2000 times smaller than that of the proton and neutron. Also note that the mass of the proton and neutron is close to 1 amu. This is a useful fact to

remember. If the number of electrons does not equal the number of protons in the nucleus then the atom is an ion:

cation: number of electrons < number of protons

anion: number of electrons > number of protons

1.2 Atomic Excitation and De-excitation

Atoms can make transitions between the orbits allowed by quantum mechanics by absorbing or emitting exactly the energy difference between the orbits. The following figure shows an atomic excitation caused by absorption of a photon and an atomic de-excitation caused by emission of a photon.

Figure 1.2: Excitation by absorption of light and de-excitation by emission of light

In each case the wavelength of the emitted or absorbed light is exactly such that the photon carries the energy difference between the two orbits. This energy may be calculated by dividing the product of the Planck constant and the speed of light hc by the wavelength of the light). Thus, an atom can absorb or emit only certain discrete wavelengths (or equivalently, frequencies or energies).

1.3 Atomic Spectra

After an atom absorbs a quantum of energy, it is said to be in an excited state relative to its normal (ground) state. When an excited atom returns to the ground state, it emits light. For example, the yellow light observed when glass is heated in a flame is due to excited sodium atoms in the glass returning to their ground state. Similarly, the familiar red light of neon signs is due to neon atoms which have been excited by an electrical discharge. When light from excited

2

atoms is viewed through a spectroscope, images of the slit appear along the scale of the instrument as a series of coloured lines. The various colours correspond to light of definite wavelengths, and the series of lines is called a line spectrum. The line spectrum of each element is so characteristic of that element that its spectrum may be used to identify it.

The simplest spectrum is that of hydrogen, the simplest element.

Figure 1.3: Balmer series of hydrogen atom; spectral lines in the visible

The part of the hydrogen spectrum which appears as visible light is shown in Figures 1.3. It should be noted that the lines at shorter wavelengths have progressively lower intensities. The wavelengths of successive lines are closer and closer together until they finally become a continuum, a region of continuous faint light. In 1885, J.J. Balmer suggested that the lines in the visible spectrum of hydrogen could be represented by the equation

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n^2} \right)$$

Where $\frac{1}{\lambda}$ is the wavenumber (inverse of wavelength), n_1 has the value 2, n_2 is an integer having the values 3, 4, 5, ∞ . The value of the constant R is $\sim 109,677.581 \text{ cm}^{-1}$. It is known as the Rydberg constant.

1.4 The Bohr Theory

Although the Balmer equation successfully represented the hydrogen spectrum, there was no theoretical justification for it. In 1914, Niels Bohr proposed a theory of the hydrogen atom which explained the origin of its spectrum and which also led to an entirely new concept of atomic structure. The Bohr model of the hydrogen atom was based on four postulates.

Postulate 1. The hydrogen atom consists of a nucleus containing a proton (and therefore having a charge of $+e$), and an electron (with a charge of $-e$) moving about the nucleus in a circular orbit of radius r (Figure 1.4).

3

Figure 1.4: The force of attraction keeps bending the path of the electron toward the nucleus and away from the straight-line motion. The net result is a circular path.

According to Coulomb's law, the force of attraction between the nucleus and the electron is

$$F = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

Where $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ (ϵ_0 is the permittivity of the vacuum; $Z = 1$ for the hydrogen atom)

4

This force just balances the centrifugal force on the electron, $\frac{mv^2}{r}$, where m is the mass of the electron and v its velocity:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r}$$

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{4\pi\epsilon_0 m v^2}{e^2}$$

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{4\pi\epsilon_0 m v^2}{e^2}$$

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$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{4\pi\epsilon_0 m v^2}{e^2}$$

The kinetic energy of the electron may be determined from this equation also:

$$\frac{1}{2} m v^2$$

$$\text{Kinetic energy} = \frac{1}{2} m v^2 = \frac{8\pi^2 m e^4}{h^2 n^2}$$

$$\frac{1}{2} m v^2 = \frac{8\pi^2 m e^4}{h^2 n^2}$$

Note:

m_e = Mass of an electron ($9.109 \times 10^{-31} \text{ kg}$)

e = Quantity of electrical charge ($1.602 \times 10^{-19} \text{ C}$)

h = Planck's constant ($6.626 \times 10^{-34} \text{ Js}$)

ϵ_0 = Vacuum permittivity ($8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

c = Speed of light ($3.0 \times 10^8 \text{ m s}^{-1}$)

R = Rydberg's constant ($1.097 \times 10^7 \text{ m}^{-1}$)

$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Postulate 2. Not all circular orbits are permitted for the electron. Only the orbits which have h

angular momentum of the electron, mvr , equal to integral multiples of 2π are allowed:

$$n h$$

$$mvr = 2\pi n \quad n = 1, 2, 3, 4, 5, \dots$$

$$n h$$

$$=$$

$$2\pi n$$

$$4$$

$$2\pi n h$$

$$2\pi n = 4\pi n^2$$

Equating the two expressions for 2π yields an expression for r , the radius of an allowed orbit:

$$2\pi n h = 2\pi n$$

$$=$$

$$n^2$$

The value $n = 1$ defines the first (smallest) orbit. Larger orbits have higher values for n .

Postulate 3. As a consequence of the restrictions on the angular momentum of an orbit, the energy of an electron in a given orbit is fixed. As long as the electron stays in that orbit, it neither absorbs nor radiates energy (hence the orbit is called a stationary state). The total energy, E , of

$$- \frac{1}{n^2}$$

an electron is the sum of its potential energy, (negative because the electric force between

$$4\pi\epsilon_0 r^2$$

$$1/n^2$$

the electron and the nucleus is attractive) and its kinetic energy, $\frac{1}{2}mv^2 =$ (from postulate

$$2\pi n h$$

1). Therefore

$$1/n^2$$

$$= -$$

$$8\pi\epsilon_0 r^2$$

Thus the total energy is equal to half the potential energy.

Substituting the expression for r from the above yields

$$n^4$$

$$= -8$$

$$2\pi n h^2$$

All of the quantities on the right-hand side of this equation are known constants except for the arbitrary integer n . Hence the possible energies of the electron are determined by the values of n .

It should be noted that the negative sign in the energy expression means that the larger the value of n , the higher will be the energy of the electron.

Postulate 4. To change from one orbit to another, the electron must absorb or emit a quantity of energy exactly equal to the difference in energy between the two orbits. When light energy is involved, the photon has a frequency given by

$$h\nu = E_2 - E_1$$

Substitution of the corresponding energy expression yields

$$\begin{aligned}
 h\nu &= 4.1 \times 10^{-18} \text{ J} \\
 \nu_2 - \nu_1 &= h\nu = (4.1 \times 10^{-18} \text{ J}) \\
 \nu &= 8.2 \times 10^{14} \text{ s}^{-1} \\
 \lambda &= 3.6 \times 10^{-7} \text{ m} \\
 &= 360 \text{ nm} \\
 &= (4^2 - 2^2) \\
 \lambda &= 8.2 \times 10^{-7} \text{ m} \\
 &= 820 \text{ nm} \\
 &= (2^2 - 1^2) \\
 &= 1.2 \times 10^{-6} \text{ m}
 \end{aligned}$$

The last equation is identical to the Balmer equation. The constant R may be evaluated by using the numerical values of ν , m , e , c , and h , and the result agrees closely with the experimentally determined value for the Rydberg constant.

It follows from the Bohr Theory, since n_1 can have values 1, 2, 3, 4, 5, etc., that several other series of line should exist in the hydrogen spectrum besides the Balmer series. Subsequent to the

theory, other series have been discovered in the ultraviolet, infrared, and far infrared regions of the spectrum. All of these series have lines with wavelengths which are given by the Balmer equation when appropriate values of n_1 and n_2 are used.

Each spectral line represents an energy difference between two possible states of the atom. Each of these states corresponds to the electron in the hydrogen atom being in an "orbit" whose radius increases with the quantum number n . The lowest allowed value of n is 1; because the electron is as close to the nucleus as it can get, the energy of the system has its minimum (most negative) value. This is the "normal" (most stable) state of the hydrogen atom, and is called the ground state.

If a hydrogen atom absorbs radiation whose energy corresponds to the difference between that of $n=1$ and some higher value of n , the atom is said to be in an excited state. Excited states are unstable and quickly decay to the ground state, but not always in a single step. For example, if the electron is initially promoted to the $n = 3$ state, it can decay either to the ground state or to the $n=2$ state, which then decays to $n = 1$. Thus this single $n = 1 \rightarrow 3$ excitation can result in the three emission lines depicted in the diagram above, corresponding to $n = 3 \rightarrow 1$, $n = 3 \rightarrow 2$, and $n = 2 \rightarrow 1$.

If, instead, enough energy is supplied to the atom to completely remove the electron, we end up with a hydrogen ion and an electron. When these two particles recombine ($\text{H}^+ + \text{e}^- \rightarrow \text{H}$), the electron can initially find itself in a state corresponding to any value of n , leading to the emission of many lines.

The lines of the hydrogen spectrum can be organized into different series according to the value of n at which the emission terminates (or at which absorption originates.) The first few series are named after their discoverers. The most well-known (and first-observed) of these is the Balmer series, which lies mostly in the visible region of the spectrum. The Lyman lines are in the ultraviolet, while the other series lie in the infrared. The lines in each series crowd together as

they converge toward the series limit which corresponds to ionization of the atom and is observed as the beginning of the continuum emission.

Figure 1.5: Line series in the hydrogen spectrum

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Table 1.2: Description of the line series of the hydrogen spectrum

Series name n_1 n_2 values Spectral region

Lyman 1 2 to ∞ Ultraviolet

Balmer 2 3 to ∞ Visible

Paschen 3 4 to ∞ Infrared

Brackett 4 5 to ∞ Far infrared

Pfund 5 6 to ∞ Far infrared

The Bohr Theory accounts for the hydrogen spectrum in the following manner. When hydrogen atoms are excited, their electrons occupy orbitals having higher energies. When an electron returns to a more stable orbit, it emits a photon of energy corresponding to the energy difference between the orbits. The transition back to the ground state can occur directly or stepwise, yielding one photon or several. The various spectral series correspond to transitions between higher orbits and those having a given value of n . It is customary to refer to the integer n as a quantum number.

The Bohr Theory can be applied with equal success to “hydrogen-like atoms”: ions containing only one electron (He^+ , Li^{2+} , etc.). The expression for the energy of the electron in the n th orbit for these ions is

$$E_n = -\frac{Z^2}{n^2} \times 13.6 \text{ eV}$$

$$E_n = -\frac{8\pi^2 m_e e^4 Z^2}{n^2 h^2}$$

Where

Where Z represents the atomic number.

1.5 Shortcomings of the Bohr Theory

1. The Bohr theory cannot explain the spectra of atoms containing more than one electron
2. Even for the atoms with one electron, the theory does not predict the relative intensities of the lines or the splitting of the lines observed when the atoms are excited in a magnetic field (the Zeeman effect). Even in the absence of external fields, the spectral lines were found to be more complex when examined with high resolution equipments. (The theory could not explain the fine details of the spectra).
3. It ignores the wave nature of the electron.
4. It violates the Heisenberg's Uncertainty Principle – Bohr claims it is possible to know exactly the position and study the motion of the electron at the same time.
5. It regards the electron as being stationary, with a specific position and distance from the nucleus.
6. It does not explain molecular bonds
7. It does not predict the relative intensities of spectral lines
8. The Bohr Model does not explain fine structure and hyperfine structure in spectral lines.

2.0 CONCLUSION

Niels Bohr proposed the Bohr Model of the Atom in 1915. Because the Bohr Model is a modification of the earlier Rutherford Model, some people call Bohr's Model the

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Rutherford-Bohr Model. The modern model of the atom is based on quantum mechanics. The Bohr Model contains some errors, but it is important because it describes most of the accepted features of atomic theory without the entire high-level math of the modern version. Unlike earlier models, the Bohr Model explains the Rydberg formula for the spectral emission lines of atomic hydrogen.

3.0 SUMMARY

The Bohr model described the energy in atoms as being quantized. Bohr explained how he perceived elements' spectral-lines; electron jumps - from high energy levels to low energy levels and thus science was able to explain phenomenon that until 1913 were not explainable.

When an electron changes to a different orbit, it either gets more energy, or it loses some energy - the larger the orbit is, the more energy. If an electron moves inward towards the core, then it emits or loses energy, while it absorbs energy when moving to an outward orbit.

This model of the atom is ideal for people who do not understand what an atom is, as of yet. It is very simple, but as things get more complicated, such as multielectron systems, this model will not be sufficient. This model of the atom has now been replaced with quantum mechanics.

4.0 TUTOR-MARKED ASSIGNMENT

Questions

1. Using the Balmer equation, find the frequency of the radiation corresponding to $n = 3$.
2. What is the frequency of the spectral line produced when an electron moves from $n = 5$ to $n = 2$ in a Hydrogen atom?
3. What value of n does the line at 656.3 nm in the Balmer series correspond to?
4. A photon with a wavelength of 397 nm is emitted from an electron in energy level 7 of a Hydrogen atom. What is the new energy level of the electron?
5. Find the frequency in Hertz of radiation with energy of 2.179×10^{-18} J per photon.
6. What frequency of light would be needed to make an electron in a Hydrogen atom jump from $n = 1$ to $n = 3$?
7. A spectral line is measured to have a wavelength of 1000 nm. Is this within the Balmer series?
8. Calculate the radius of the first allowed Bohr orbit for hydrogen
9. Calculate the energy of an electron in the first Bohr orbit of hydrogen
10. Calculate the wavelength of the first line and the series limit for the Lyman series for hydrogen

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THE QUANTUM MECHANICAL MODEL OF THE ATOM

INTRODUCTION

The Bohr Theory was a milestone; however, it had major defects. It explained the energies of the

electron transitions in hydrogen and hydrogen-like atoms, but not in any atom having more than one electron. Even for hydrogen, the theory could not explain the different intensities of the spectral lines.

In this unit, a more sophisticated theory of the electronic structure of the atom will be described. This theory, called wave mechanics, not only gives results as good as the Bohr Theory for the case of the hydrogen atom but also can be used to explain both the energies and the intensities of the spectral lines of hydrogen. Moreover, approximate descriptions of atoms which have more than one electron may be given in terms of this theory.

The branch of science that takes into account this dual behaviour of matter is called quantum mechanics. Quantum mechanics is a theoretical science that deals with the study of the motion of the microscopic objects that have both observable wave like and particle like properties. This science was developed independently by Louis de Broglie, Werner Heisenberg and Erwin Schrödinger.

The quantum mechanical model is based on quantum theory, which says matter also has properties associated with waves. According to quantum theory, it's impossible to know the exact position and momentum of an electron at the same time. This is known as the Uncertainty Principle.

The quantum mechanical model of the atom uses complex shapes of orbitals (sometimes called electron clouds), volumes of space in which there is likely to be an electron. So, this model is based on probability rather than certainty.

The major difference is that in the Bohr model the electrons revolve around the nucleus in fixed orbits similar to the way planets orbit around the sun. The wave mechanical model, influenced by the Heisenberg uncertainty principle, says that electrons do not orbit in fixed orbits. In fact, it's impossible to know both the position and momentum of a particle like an electron. Instead the wave mechanical model uses the Schrödinger equation to predict the probabilities of where the electron may be positioned at any given time (without saying with any certainty where the electron actually is).

There are other differences (differences in shapes of electron orbits/atomic orbitals; failure of the Bohr model to account for hybrid orbitals, molecular orbitals, resonance, etc.), but the major difference is that Bohr's model gives the electron orbit an exact travel path while the wave mechanical model doesn't claim to know where the electron is at any given time, only probabilities of where it is likely to be).

MAIN CONTENT

Energy Is Quantized (The Planck-Einstein Relation: $E = h\nu$)

After Max Planck determined that energy is released and absorbed by atoms in certain fixed amounts known as quanta, Albert Einstein took his work a step further, determining that radiant energy is also quantized—he called the discrete energy packets photons. Einstein's theory was

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that electromagnetic radiation (light, for example) has characteristics of both a wave and a stream of particles.

The Planck-Einstein relation is the equation relating energy to frequency. It was the first

equation of quantum mechanics, implying that energy comes in multiples (“quanta”) of a fundamental constant h . It is written as

$$E = h\nu$$

ν is linear frequency and ω is angular frequency. The fundamental constant h is called Planck’s constant and is equal to $6.62608 \times 10^{-34} \text{ Js}$

This relation was first proposed by Planck in 1900 to explain the properties of black body radiation.

The interpretation was that matter energy levels are quantized. At the time this appeared compatible with the notion that matter is composed of particles that oscillate. The discovery that the energy of electrons in atoms is given by discrete levels also fitted well with the Planck relation.

In 1905 Einstein proposed that the same equation should hold also for photons, in his explanation of the photoelectric effect.

The light incident on a metal plate gives rise to a current of electrons only when the frequency of the light is greater than a certain value. This value is associated with the energy required to remove an electron from the metal (the “work function”). The electron is ejected only when the light energy matches the discrete electron binding energy. Einstein’s proposal that the light energy is quantized just like the electron energy was more radical at the time: light quantization was harder for people to accept than quantization of energy levels of matter particles. (The word “photon” for these quantized packets of light energy came later, given by G. N. Lewis, of Lewis Hall!)

Wave-Particle duality

In 1924, a young French scientist, Louis de Broglie, proposed that, like light, a stream of electrons might have wave properties in addition to particle properties. He suggested that a particle such as an electron travelling at a velocity v possessed a wavelength λ which is given by

$$\lambda = \frac{h}{mv}$$

Where h is Planck’s constant and m is the mass of the particle. In 1927, two Americans, C.J. Davison and L.H. Germer, measured the wavelength of an electron beam by diffraction through a crystal in a manner similar to the diffraction of X rays. Thus de Broglie’s hypothesis was confirmed.

De Broglie proposed this relationship as a general one. With every particle, there is an associated wave. The wavelength of the particle depends on its mass and how fast it is moving.

A particle occupies a particular location, but a wave has no exact position. A wave extends over some region of space. Because of their wave properties, electrons are always spread out rather than located in one particular space. As a result, the position of an electron cannot be precisely

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defined. Electrons are said to be delocalized because their waves are spread out rather than pinpointed.

Viewing the electron and other particle-waves as delocalized, however, changes the way we look at things. Instead of things having an exact location and motion, they are distributed over some

volume.

An electrons beam is deflected by a magnetic field; this is evidence that electrons have a particle-like nature. Electron diffraction is an evidence for the wave nature of the electron.

The Uncertainty Principle

This is also known as the Indeterminacy principle. Werner Heisenberg, a German Physicist, found in the 1920s that the position and motion of a particle-wave cannot be “pinned down”. If a particular particle-wave can be pinpointed in a specific location, its motion must be unknown. Conversely, if the motion of a particular particle-wave is known precisely, its location must be unknown. Heisenberg summarized this uncertainty in what has become known as the uncertainty principle: The more accurately position is known, the less accurately can the momentum be determined, and vice-versa. Mathematically we describe the uncertainty principle as the following, where 'x' is position and 'p' is momentum:

h

$\Delta x \cdot \Delta p >$

$\frac{h}{2}$

Where h is Planck's constant. The principle, which was first stated by Heisenberg, arises from the dual particle-wave nature of matter.

For example, you can measure the location of an electron, but not its momentum (energy) at the same time.

This is perhaps the most famous equation next to $E=mc^2$ in physics. It basically says that the combination of the error in position times the error in momentum must always be greater than Planck's constant. So, you can measure the position of an electron to some accuracy, but then its momentum will be inside a very large range of values. Likewise, you can measure the momentum precisely, but then its position is unknown.

The Schrödinger equation

In 1926, Erwin Schrödinger postulated an equation similar to equations which describe wave motion, the solutions of which describe the properties of the electron in the atom. The solutions to Schrödinger equation are exact for the electron in the hydrogen atom and approximate for the

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electrons in other atoms. These solutions are related to the probability of finding the electron in a given locality in the atom.

The wave function

Since an electron has wave properties, it is described as a wave function, ψ , or $\psi(x,y,z)$; the latter meaning that ψ is a function of coordinates x , y , and z . The probability of finding an electron in any volume element in space is proportional to the square of the absolute value of the wave function, integrated over that volume of space. This is the physical significance of the wave function. Measurements we make of electronic charge density, then, should be related to $|\psi|^2$, and not $|\psi|$.

Expressed as an equation, we have

Probability(x,y,z) $\propto |\psi(x,y,z)|^2$.

The Schrödinger equation is commonly written in the form:

$$H\psi = E\psi$$

This looks deceptively simple.

H is called the Hamiltonian operator and contains terms for the kinetic and potential energies of the system.

E is the numerical value of the energy for any particular ψ .

The complete Schrödinger equation is

$$\frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} + \frac{1}{2} \frac{\partial^2 \psi}{\partial y^2} + \frac{1}{2} \frac{\partial^2 \psi}{\partial z^2} + [V - E] \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = -2h^2$$

Before the Schrödinger equation can be solved, the exact form of V must be specified. V contains information about all the Coulombic attractions and repulsions among the electrons and nuclei of the atom or molecule.

V = Potential energy.

The solutions of the Schrödinger equation involve integers which determine the energies and momenta of the electrons. The integers correspond to the quantum numbers of the Bohr theory, but in this case they are required by the mathematical form of the wave equation, whereas the quantum numbers of the Bohr theory were assumed arbitrarily. At this point it is not necessary to learn how to solve the Schrödinger equation. It is only necessary to know that the solutions are used to describe the arrangement of electron in atoms and that the quantum numbers are the most characteristic feature of the solutions. The arrangements of the electrons in atoms can be predicted in terms of the quantum numbers alone. The properties of electrons which are determined by this arrangement include their energies, their orientations in space, and their interactions with other electrons within the same atom and with electrons of other atoms. All the chemical and physical properties of an atom depend on the arrangement of its electrons; therefore the quantum numbers are immediately useful in providing chemical information. Each electron in an atom is assigned four quantum numbers, which define its energy, its orientation in space, and its possible interaction with other electrons.

Quantum Numbers

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By solving the Schrödinger equation ($H\psi = E\psi$), we obtain a set of mathematical equations, called wave functions (ψ), which describe the probability of finding electrons at certain energy levels within an atom.

A wave function for an electron in an atom is called an atomic orbital; this atomic orbital describes a region of space in which there is a high probability of finding the electron. Energy changes within an atom are the result of an electron changing from a wave pattern with one energy to a wave pattern with a different energy (usually accompanied by the absorption or emission of a photon of light).

Each electron in an atom is described by four different quantum numbers. The first three (n, l, ml) specify the particular orbital of interest, and the fourth (ms) specifies how many electrons can

occupy that orbital.

i. Principal Quantum Number (n): $n = 1, 2, 3, \dots, \infty$

Specifies the energy of an electron and the size of the orbital (the distance from the nucleus of the peak in a radial probability distribution plot). All orbitals that have the same value of n are said to be in the same shell (level). For a hydrogen atom the most stable state is when $n = 1$, and levels for $n > 1$ are excited states of the atom. The total number of orbitals for a given n value is n^2 .

ii. Angular Momentum (Azimunthal) Quantum Number (l): $l = 0, \dots, n-1$.

As the name implies, it specifies the angular momentum of the electron. In any atom other than the hydrogen atom, n and l together define the energy of the electron. l also specifies the shape of an orbital with a particular principal quantum number. The secondary quantum number divides the shells into smaller groups of orbitals called subshells (sublevels). Usually, a letter code is used to identify l to avoid confusion with n :

l 0 1 2 3 4 5

Letter s p d f g h

The subshell with $n=2$ and $l=1$ is the 2p subshell; if $n=3$ and $l=0$, it is the 3s subshell, and so on. The value of l also has a slight effect on the energy of the subshell; the energy of the subshell increases with l ($s < p < d < f$).

iii. Magnetic Quantum Number (m_l): $m_l = -l, \dots, 0, \dots, +l$.

Specifies the orientation in space of an orbital of a given energy (n) and shape (l). It defines the possible orientations of the angular momentum in space with respect to some arbitrarily defined axis. The magnetic quantum number becomes important in situations in which the electron interacts with external magnetic fields, including the fields generated by the motions of other electrons. This number divides the subshell into individual orbitals which hold the electrons; there are $2l+1$ orbitals in each subshell. Thus the s subshell has only one orbital, the p subshell has three orbitals, and so on.

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iv. Spin Quantum Number (m_s): $m_s = +\frac{1}{2}$ or $-\frac{1}{2}$.

To account for the fine details observed in atomic spectra, it is necessary to introduce the fourth quantum number, m_s , which specifies the orientation of the spin axis of an electron. This quantum number takes into account the spinning of the electron about its own axis as it moves about the nucleus. An electron can spin in only one of two directions. Although m_s has the possible non-integral values $-\frac{1}{2}$ and $+\frac{1}{2}$, the difference between the permitted values is integral.

Table of Allowed Quantum Numbers

n l m_l No. of orbitals Orbital name No. of Maximum no.

electrons of electrons

1 0 0 1 1s 2 2

2 0 0 1 2s 2 8

1 -1, 0, +1 3 2p 6

3 0 0 1 3s 2

1 -1, 0, +1 3 3p 6 18

2 -2, -1, 0, +1, +2 5 3d 10

4 0 0 1 4s 2

1 -1, 0, +1 3 4p 6 32

2 -2, -1, 0, +1, +2 5 4d 10

3 -3, -2, -1, 0, +1, +2, +3 7 4f 14

Sublevels are broken down into orbitals; an orbital holds 2 electrons

No. of electrons in No. of

sublevel Names of each orbital

each sublevel orbitals

s 2 1 s

p 6 3 pz px py

d 10 5 dz² dxz dyz dxy dx²-y²

f 14 7 fz³ fxz² fyz² fxyz fz(x²-y²) fx(x²-3y²) fy(3x²-y²)

ORBITALS

Atomic orbital – The volume containing all the points within a free atom at which the wave function of an electron has an appreciable magnitude. More simply, orbital is a region around the nucleus of an atom where the probability of finding an electron is high.

A solution of the Schrödinger equation, expressed in terms of a set of permitted values of n , l , and m_l , defines an orbital. The most characteristic property of an orbital is its energy. Since in the absence of external field the value of m_l does not influence the energy, orbitals are grouped into sets called subshells, which are denoted by the values of n and l only. The use of two

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numbers is awkward, hence orbitals with l values of 0, 1, 2, and 3, are denoted as s, p, d, and f, respectively.

The energies of atomic orbitals also describe their shapes. The shapes are uncertain, but predictions have been made by experimentation. Another difficult task is describing where an electron is. We can think of it as a wave, and describing its exact location is impossible for us to comprehend. Instead, we can think of it as the statistical probability of the electron being found at a particular place.

At any given moment, an electron can be found at any distance from the nucleus and in any direction according to the Heisenberg Uncertainty Principle.

The s orbital

The s orbital is a spherically-shaped region describing where an electron can be found, within a certain degree of probability. The shape of the orbital depends on the quantum numbers associated with an energy state. All s orbitals have $l = m = 0$, but the value of n can vary.

The p orbital

The p orbital is a dumbbell-shaped region describing where an electron can be found, within a certain degree of probability. The shape of the orbital depends on the quantum numbers associated with an energy state. Since there are 3 p orbitals per energy level, the lobes extend out

along the x-axis (px orbital), the y-axis (py orbital), and the z-axis (pz orbital).

px orbital py orbital pz orbital

The d orbital

The d orbital's shapes are even more complex because there are 5 orbitals in a d subshell. Four of the five d orbitals (dxy, dxz, dyz, and dx²-y²) have four lobes extending out perpendicular to each

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other. The last one, dz², has two lobes extending out along the z-axis with a torus (doughnut-shaped ring) around the center on the x-y plane.

dxy orbital dxz orbital dyz orbital dx²-y² orbital dz² orbital

The energy of an electron in a given orbital can be specified precisely from the values of its quantum numbers. As a consequence, the position of the electron is quite uncertain. Therefore it turns out that while the solutions to the Schrödinger equation give the energies of the electron, they can give only the probability of finding the electron in any arbitrary volume around the nucleus.

The probability distribution for an electron in a 1s, 2s or 3s orbital of hydrogen is:

Radial distribution functions of the 1s and 2s orbitals.

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Note that as the principal quantum number increases, the maximum in the probability density occurs at larger distances for orbitals of the same angular momentum type. For example, the maximum probability for the 1s orbital occurs at 1 bohr (0.529 Å), for the 2s orbital the primary maximum occurs at 5.3 bohr, and for the 3s orbital the primary maximum occurs at about 13 bohr.

For the hydrogen atom, it is seen that there is maximum probability of finding the electron at a distance of 1 bohr (0.529 Å) from the nucleus. This distance is the same as the experimental radius of the hydrogen atom and also agrees with the radius derived by means of the Bohr theory. Similar representation for the 2s orbital of hydrogen is shown – there is a greater probability of finding the electron further away from the nucleus than in the case of the 1s orbital, but there is still some chance of finding the electron very close to the nucleus. Also significant is the fact that at an intermediate distance, there is a surface at which the probability of finding the electron is zero. Surfaces at which the probability goes to zero are called nodes. For any orbital having principal quantum number n, there are always n-1 nodes. This nodal character of the probability distribution is consistent with the assumption that the motion of the electron has the character of a wave.

Writing Electron Configurations

The Pauli Exclusion Principle

The Pauli Exclusion Principle (Wolfgang Pauli, Nobel Prize 1945) states that no two electrons in the same atom can have identical values for all four of their quantum numbers. What this means is that no more than two electrons can occupy the same orbital, and that two electrons in the same orbital must have opposite spins.

Because an electron spins, it creates a magnetic field, which can be oriented in one of two directions. For two electrons in the same orbital, the spins must be opposite to each other; the spins are said to be paired. These substances are not attracted to magnets and are said to be diamagnetic. Atoms with more electrons that spin in one direction than another contain unpaired electrons. These substances are weakly attracted to magnets and are said to be paramagnetic.

3.7.1. The Aufbau Principle

The distribution of electrons among the orbitals of an atom is called the electron configuration. The electrons are filled in according to a scheme known as the Aufbau principle ("building-up"), which corresponds (for the most part) to increasing energy of the subshells:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f

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It is not necessary to memorize this listing, because the order in which the electrons are filled in can be read from the periodic table in the following fashion:

Or, to summarize:

In electron configurations, write in the orbitals that are occupied by electrons, followed by a superscript to indicate how many electrons are in the set of orbitals (e.g., H 1s¹).

Another way to indicate the placement of electrons is an orbital diagram, in which each orbital is represented by a square (or circle), and the electrons as arrows pointing up or down (indicating the electron spin).

3.7.2. The Hund's Rule of Maximum Multiplicity

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When electrons are placed in a set of orbitals of equal energy, they are spread out as much as possible to give as few paired electrons as possible (Hund's rule).

According to this rule, electron pairing will not take place in orbitals of same energy (same subshell) until each orbital is first singly filled with parallel spin. In other words in a set of orbitals having same energy (degenerate orbitals), the electrons distribute themselves to occupy separate orbitals with same spin as far as possible. This rule can be illustrated by considering the example of carbon. The atomic number of carbon is 6 and it contains two electrons in 2p subshell and these can be distributed in the following three ways:

Since all the three 2p orbitals have same energy, therefore, it does not take any difference as to which of the three orbitals contain electrons. In state (a) both the electrons are in the same orbital. In state (b), the two electrons are present in different orbitals but with opposite spins while in state (c), the electrons are present in different orbitals with same spins. Now, the electrons are charged particles and repel one another. The electron-electron repulsions are minimum when the electrons are as far apart as possible with parallel spins. Thus, state (c) has minimum repulsions and corresponds to lower energy (stable state). This is in accordance with Hund's rule. This principle is very important in guiding the filling of p, d and f subshells, which have more than one type of orbitals.

In a ground state configuration, all of the electrons are in as low an energy level as it is possible for them to be. When an electron absorbs energy, it occupies a higher energy orbital, and is said to be in an excited state.

4.0. SUMMARY

In this Unit, we observed that electrons do not quite behave that nicely. Instead, electrons behave like a particle AND a wave, so a new model was created. We discussed the differences between the Bohr model and the quantum mechanical model, the similarities and how we can represent/illustrate electron locations via electron configuration and box diagrams. We also learned that solution to the Schrödinger equation gave rise to the quantum numbers, which describe the behaviour of electrons in atoms. The different types of orbitals and rules governing the writing of electronic configurations of atoms were also were discussed.

5.0. TUTOR-MARKED ASSIGNMENTS

1. What is the de Broglie wavelength of a person with a mass of 50 kg jogging at 5 m/s?
2. What is the de Broglie wavelength of an electron moving at 2.2×10^6 m/s?
3. Electrons experience a drop in energy of 6.409×10^{-15} J, what is the wavelength of the electrons?

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